

AMAN DIXIT

AI/ML Engineer | Agentic AI Developer | Python & FastAPI Specialist | GenAI Solutions

Bengaluru, Karnataka, India | +91-8595604117 | dixitaman.now.wwe@gmail.com | linkedin.com/in/aman-dixit-la999117a

6+ Years Building AI & Full-Stack Systems | 4 Agentic AI Systems Shipped | 3 LLM Platforms Integrated | 50K+ Daily AI API Requests Handled | 60% Incident Reduction via AI Automation

PROFESSIONAL SUMMARY

AI/ML Engineer and Agentic AI Developer with 5+ years of experience building intelligent, production-grade AI systems using Python, FastAPI, LangChain, OpenAI GPT, Claude API, and RAG architectures. Deep expertise in designing and deploying Agentic AI workflows — including multi-agent orchestration, tool-use pipelines, MCP (Model Context Protocol), and context-aware LLM applications. Strong full-stack engineering foundation (MERN Stack: React.js, Node.js, MongoDB) enables end-to-end ownership from AI backend design to frontend integration. Proven track record of integrating GenAI into healthcare, SaaS, and enterprise platforms — delivering measurable outcomes: 60% reduction in incidents, 35% latency improvement, and 50,000+ daily AI-powered API requests at 99.5% uptime. Passionate about pushing the frontier of Agentic AI, Retrieval-Augmented Generation, and LLM-powered automation.

TECHNICAL SKILLS

AI & GenAI (Core): Python | FastAPI | LangChain | LlamaIndex | OpenAI GPT (3.5/4/4o) | Claude API (Anthropic) | Gemini API | Prompt Engineering | RAG (Retrieval-Augmented Generation) | Vector Databases | Embeddings

Agentic AI & Orchestration: Agentic AI Workflows | Multi-Agent Systems | MCP (Model Context Protocol) | Tool-Use Pipelines | ReAct Agents | Function Calling | Agent Memory & Context Management | CrewAI | AutoGen | LangGraph

AI Infrastructure: FastAPI | Python Microservices | Async Python (asyncio) | Celery | Redis | WebSockets | REST API Design | Docker | AWS Lambda | GCP | Nginx | CI/CD (Jenkins)

Vector & Data: Pinecone | ChromaDB | FAISS | Weaviate | PostgreSQL (pgvector) | MongoDB | MySQL | Redis | Data Pipelines | ETL | Chunking Strategies | Embedding Models (OpenAI, HuggingFace)

ML & NLP: HuggingFace Transformers | Sentence Transformers | Fine-tuning | NLP Pipelines | Text Classification | Semantic Search | Document Intelligence | OCR Integration

Full-Stack (Supporting): React.js | Node.js | Express.js | TypeScript | MongoDB | REST APIs | JWT Auth | WebSockets | HTML5 | CSS3

DevOps & Cloud: Docker | Jenkins | CI/CD | AWS (Lambda, S3, EC2) | GCP | Git | GitHub | Linux | Nginx | Monitoring & Logging

Tools & Methodologies: Jupyter Notebooks | VS Code | Postman | Jira | Agile/Scrum | LLMOps | Prompt Versioning | A/B Testing for AI

PROFESSIONAL EXPERIENCE

Senior AI/ML Engineer & Agentic AI Developer

07/2024 – 03/2026

Advantage AI Engineering Private Limited | Bengaluru, India

Owned AI/backend architecture and Agentic AI delivery for 3 HIPAA-compliant healthcare platforms, integrating LLMs, RAG pipelines, and Python microservices with a React.js/Node.js full-stack.

- Architected and deployed 4 Agentic AI systems using Python, FastAPI, and LangChain — including a medical Q&A agent, a document intelligence agent, a symptom analysis agent, and a clinical summarization pipeline — handling 50,000+ daily AI API requests at 99.5% uptime.
- Built production RAG pipelines using LangChain and LlamaIndex with ChromaDB and FAISS vector stores, enabling intelligent document retrieval over 100,000+ healthcare records with sub-2-second query response times.
- Designed and implemented MCP (Model Context Protocol) integration enabling dynamic tool-use, context persistence, and multi-step reasoning across Agentic AI sessions — reducing hallucination rates by 40%.

- Developed FastAPI microservices for LLM orchestration, async inference endpoints, streaming responses (Server-Sent Events), and function-calling pipelines — integrated with React.js frontend via WebSockets and REST APIs.
- Engineered multi-agent orchestration workflows using CrewAI and LangGraph — coordinating specialized agents (retrieval, reasoning, summarization, output formatting) for complex clinical workflows.
- Integrated OpenAI GPT-4o and Claude API (Anthropic) with structured Prompt Engineering techniques (chain-of-thought, few-shot, system prompts, output parsers) — achieving 92% task completion accuracy on healthcare Q&A benchmarks.
- Built Python-based automated security scanning and AI output validation pipelines, achieving a 60% reduction in critical production incidents and ensuring compliance with HIPAA data handling standards.
- Implemented LLMOps practices: prompt versioning, A/B testing of LLM configurations, latency monitoring, cost tracking per token, and automated regression testing for AI outputs.
- Collaborated with React.js frontend team to design AI-first UX patterns — streaming chat interfaces, real-time agent status indicators, and source citation displays for RAG responses.
- Reduced AI inference costs by 30% through intelligent caching (Redis), prompt compression, and model routing strategies (directing simple queries to GPT-3.5, complex to GPT-4o).

Senior Software Engineer (Full-Stack + AI Integration)

09/2023 – 04/2024

NIIT | Bengaluru, India

Full-stack development on a SaaS learning platform, introducing initial AI-powered features using Python and OpenAI API alongside the core Node.js/React.js stack.

- Integrated OpenAI GPT-3.5 API into a learning platform to power AI-generated quiz questions, content summaries, and personalized learning path recommendations — improving learner engagement by 28%.
- Built Python Flask microservices for AI-powered content processing: text extraction, semantic chunking, and embedding generation for a basic RAG-based course search feature.
- Developed Node.js/Express.js REST APIs and React.js frontend components supporting AI feature delivery — owning the full integration stack from Python AI backend to browser UI.
- Implemented async Python processing pipelines using Celery and Redis for background AI inference jobs, preventing UI blocking on long-running LLM tasks.
- Introduced prompt engineering standards for the team: structured templates, output format instructions, and fallback strategies — reducing AI response errors by 35%.
- Mentored 3 junior developers on AI integration patterns, FastAPI best practices, and LLM API usage — establishing the team's first AI engineering playbook.

Senior Software Developer (Full-Stack)

12/2021 – 06/2023

Ksolves India Limited | Bengaluru, India

Full-stack SaaS product development. Built strong Python scripting and automation foundations used in later AI engineering roles.

- Developed and maintained 12+ SaaS product features using React.js, Node.js, Express.js, and MongoDB — serving 10,000+ monthly active users with 99%+ availability.
- Built Python automation scripts for data migration, ETL pipelines, and report generation — processing 500,000+ records with scheduled Celery jobs and Redis queuing.
- Integrated third-party REST APIs and designed scalable Node.js backend services with JWT authentication, WebSocket real-time features, and GraphQL endpoints.
- Implemented automated test suites with Jest and Mocha achieving 80%+ coverage, and Next.js SSR for improved SEO and performance — reducing page load times by 30%.

Software Developer

06/2020 – 11/2021

LitSpark Solutions | Bengaluru, India

- Delivered 6 full-stack web applications using JavaScript, React.js, Node.js, PHP, and MySQL for education and retail clients — building foundational full-stack engineering skills.
- Wrote Python utility scripts for data processing, automated reporting, and system integrations — beginning Python engineering journey that evolved into AI/ML specialization.

Software Engineer

06/2019 – 04/2020

Get Ahead Education Limited | Bengaluru, India

- Built enterprise HR automation system in PHP, MySQL, and JavaScript, reducing manual effort by 80% and processing 1,000+ student registrations with automated workflows.

AI / GENAI PROJECTS

MedAgent — Multi-Agent Clinical AI Assistant | Tech: Python | FastAPI | LangGraph | OpenAI GPT-4o | ChromaDB | RAG | MCP | React.js | WebSockets

- Designed and built a multi-agent AI system for healthcare providers — coordinating 4 specialized agents (Retrieval Agent, Symptom Analyzer, Clinical Summarizer, Response Formatter) using LangGraph orchestration and MCP for context persistence.
- RAG pipeline ingested 100,000+ medical documents into ChromaDB with HuggingFace embeddings, enabling sub-2-second semantic retrieval with source citation in responses.
- FastAPI async backend served streaming LLM responses via Server-Sent Events to React.js chat UI — delivering real-time, token-by-token output with agent status visibility.
- Achieved 92% accuracy on clinical Q&A benchmark, 40% hallucination reduction vs. baseline, and 30% cost savings via model routing and Redis caching.

DocuMind — Intelligent Document Processing & Extraction Agent | Tech: Python | FastAPI | LangChain | Claude API | FAISS | Celery | Redis | PostgreSQL

- Built an Agentic AI pipeline for automated document intelligence: PDF ingestion, OCR, semantic chunking, vector embedding (FAISS), and LLM-powered structured data extraction.
- Agent used function-calling and ReAct reasoning to extract entities, dates, amounts, and relationships from unstructured medical forms and insurance documents with 89% extraction accuracy.
- FastAPI async endpoints handled concurrent document processing with Celery background workers and Redis task queuing — processing 500+ documents per hour at scale.
- Integrated Claude API (Anthropic) for high-accuracy document summarization with structured JSON output using Prompt Engineering and output parsers.

AutoLearn AI — Personalized Learning Path Generator | Tech: Python | FastAPI | OpenAI GPT-3.5/4 | LangChain | Pinecone | Node.js | React.js

- Designed a GenAI-powered SaaS feature that generates personalized learning paths by analyzing learner performance history, goals, and skill gaps using LangChain agents and OpenAI GPT-4.
- Built vector-based course recommendation engine with Pinecone — embedding course content and learner profiles to enable semantic similarity matching with 85% learner satisfaction score.
- FastAPI backend served recommendation API endpoints consumed by React.js frontend; Node.js handled authentication, user management, and progress tracking.
- Implemented prompt engineering techniques (few-shot, chain-of-thought, structured output) to generate curriculum-aligned learning plans with consistent, parseable JSON responses.

AgentOps Monitor — LLMops Observability Dashboard | Tech: Python | FastAPI | Prometheus | Grafana | Redis | React.js

- Built an internal LLMops observability tool to track AI agent performance: token usage, latency per agent step, prompt version comparisons, error rates, and cost per session.
- FastAPI backend collected telemetry from all production AI agents; Prometheus scraped metrics and Grafana dashboards visualized agent health in real time.
- Enabled data-driven prompt optimization: A/B tested 12 prompt variants, identifying configurations that improved response quality by 25% and reduced average tokens-per-request by 18%.

EDUCATION

Bachelor of Computer Applications (BCA)

04/2020

IFIM College, Bengaluru

CERTIFICATIONS

Google Project Management Professional Certificate

Coursera / Google

Agile Project Management | Sprint Planning | Stakeholder Management | Risk Governance

MERN Full-Stack Development Certification

Udemy

Full-Stack Foundation: MongoDB | Express.js | React.js | Node.js — enables end-to-end AI system ownership